

RE-GREENING THE HEARTLAND

A 50 Million Acre Prairie Restoration Initiative
Risk Insulation • Water Security • Strategic Autonomy

Second Edition — April 2026

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Executive Summary

The Re-Greening the Heartland Initiative proposes a transformative 50 million acre prairie restoration project across the Great Plains states. It is designed to achieve three mutually reinforcing objectives: insulating farm households from the input-cost volatility and commodity-price fragility that now define dryland agriculture; restoring ecological function at landscape scale, including measurable improvements to water security in critical aquifer regions; and strengthening national strategic autonomy by reducing American agriculture's dependence on foreign-sourced fertilizer, fuel, and commodity-export infrastructure.

This second edition, updated in April 2026, reflects a policy and economic landscape that has changed substantially since the original proposal in May 2025. Fertilizer prices have spiked nearly 46% in a single month due to conflict-driven disruption in the Strait of Hormuz. Diesel costs have surged from \$3.52 to \$4.92 per gallon. The USDA has lost over 24,000 employees—a 27% workforce reduction—including 22% of Natural Resources Conservation Service staff, gutting the federal agency most responsible for conservation program delivery. The 2018 Farm Bill remains on its third extension with no comprehensive successor. And the Ogallala Aquifer continues to decline, with southwest Kansas losing over 1.5 feet of water table in a single year and a new state law requiring groundwater management districts to submit action plans by July 2026.

These developments have made the original proposal's core argument—that stable stewardship income on restored prairie is economically superior to volatile crop production on marginal dryland acres—substantially stronger. They have also clarified that the initiative's governance cannot depend on a functioning federal conservation bureaucracy. This edition therefore proposes a state-chartered, farmer-owned Prairie Restoration Cooperative with independent technical capacity, funded through a layered model that aggregates federal conservation payments, state water-security appropriations, bond finance, and ecosystem-service market revenue.

1. Why This Matters Now: Geopolitics Has Made Input-Intensive Agriculture More Fragile

The American heartland faces a convergence of pressures that have intensified since the original edition of this proposal:

Fertilizer chokepoint exposure. The conflict centered on Iran and the Strait of Hormuz has disrupted global fertilizer supply chains. The World Bank reports urea prices surged nearly 46% between February and March 2026. Reuters reports fertilizer prices at New Orleans jumped from \$516/metric ton to as much as \$683 as the war disrupted supply. FAO has issued formal warnings about risks to global agrifood systems from Hormuz disruption. American producers who depend on imported potash, phosphate intermediates, and natural-gas-derived nitrogen are directly exposed.

Fuel price volatility. EIA data show on-highway diesel rising from \$3.52/gallon in January 2026 to \$4.92/gallon by late March—a 40% increase that flows directly into every field operation from planting to harvest to grain transport.

Commodity oversupply and weak pricing power. USDA's April 2026 wheat outlook shows rising global stocks and continued export competition from the Black Sea region, Brazil, and Argentina. U.S. producers face persistent margin compression. Kansas State University's 2025 wheat budgets showed total costs exceeding returns under middle-yield scenarios, with weighted average total cost of production at \$6.91/bushel against marketing-year average prices around \$5.50–\$5.90/bushel.

Ogallala Aquifer depletion. Southwest Kansas saw aquifer levels decline 1.52 feet in 2024 alone—accelerating from the prior year. A University of Texas projection indicates up to 70% of the Texas Panhandle's Ogallala may become unusable within 20 years at current pumping rates. Kansas now legally requires all five groundwater management districts to submit action plans by July 1, 2026. Farmers in western Kansas have already abandoned over half their wells in some districts.

Federal conservation infrastructure degradation. USDA lost over 24,000 employees in 2025—a 27% workforce reduction driven by DOGE-initiated deferred resignation programs. NRCS lost 22% of its staff nationwide, with Kansas losing 32% of all USDA employees. Many of the departed staff were mid-career specialists with 12–15 years of experience in soil science, hydrology, and conservation planning. Historic conservation funding from the Inflation Reduction Act and reconciliation packages cannot be effectively delivered because the staff who administer these programs no longer exist.

This convergence creates a paradox: there has never been more federal money available for conservation, and never fewer people to deliver it. The Re-Greening the Heartland Initiative addresses this paradox by proposing a governance structure that builds independent technical capacity rather than depending on federal staffing levels that have proven volatile across administrations.

2. From Crops to Prairie Stewardship: The Farmer Financial Model

2.1 The Current Agricultural Reality

The economic case for prairie transition has strengthened considerably since the original proposal. Farmers on marginal dryland acres in the central Great Plains face a triple squeeze: weak commodity prices that fail to cover total production costs, rising input costs driven by geopolitical disruption to fertilizer and fuel supply chains, and declining water availability that threatens the viability of irrigated operations.

The point of the program is not that prairie will always beat every crop acre in every year. It is that prairie stewardship can dominate marginal dryland acres on a risk-adjusted basis under repeated

high-input and weak-price regimes—which is precisely the regime American wheat producers now inhabit.

2.2 Scenario Analysis: Conventional Wheat vs. Prairie Stewardship

Rather than presenting a single static comparison, this edition models three scenarios that reflect the range of conditions dryland wheat producers actually face. All figures are per acre for central Great Plains dryland wheat, based on Kansas State University budget data and USDA market reports.

	Baseline Year	Input Shock	Price Squeeze	Prairie Stewardship
Wheat price (\$/bu)	\$5.90	\$5.90	\$4.80	n/a
Yield (bu/acre)	40	40	32	n/a
Gross revenue	\$236	\$236	\$154	\$225 (guaranteed)
Fertilizer + chemicals	\$95	\$139 (+46%)	\$95	\$0
Fuel + equipment	\$58	\$81 (+40%)	\$58	\$5–10
Labor	\$47	\$47	\$47	\$15–25
NET INCOME	\$36	–\$31	–\$46	\$190–205
Financial risk	Moderate	Severe	Severe	Low

Under baseline conditions, dryland wheat returns approximately \$36/acre—marginal profitability that one bad weather event can erase. Under an input-cost shock (fertilizer +46%, diesel +40%—both of which occurred in early 2026), the same acre produces a net loss of \$31. Under a combined price-squeeze and yield-reduction scenario (drought year with weak markets), losses deepen to \$46/acre. Prairie stewardship, by contrast, delivers \$190–205/acre in guaranteed net income across all three scenarios because it has no input costs, no fuel dependence, and no commodity-price exposure.

Participating farmers also retain access to supplementary income streams that are compatible with restored prairie: managed grazing (\$15–40/acre), native seed harvesting (\$10–25/acre), hunting leases (\$5–20/acre), and agritourism revenue. These are additive to the base stewardship payment and diversify income further.

2.3 The Transition Journey

When a farmer enrolls, the process unfolds in four stages:

1. Assessment and planning (months 1–3): enrollment in the Cooperative, property-specific restoration plan, baseline ecological and carbon assessment, contract finalization.
2. Transition year (year 1): final conventional harvest on enrolled acres, site preparation using no-till approaches, initial native seed planting, first annual payments begin.

3. Establishment phase (years 2–4): minimal management focusing on establishment, periodic mowing for weed control, monitoring and adaptive management, full annual payments continue.
4. Mature management (years 5+): shift to prescribed fire and managed grazing regimes, performance-based payment bonuses, seed harvesting and ecotourism development.

3. Acreage Classification: Three Classes of Enrollment

Not all 50 million acres serve the same function. This edition introduces a three-class enrollment framework that improves the proposal's economics, politics, and hydrological precision simultaneously.

3.1 Class A: Marginal Dryland Cropland Retirement

These are the least profitable acres in the Great Plains—dryland wheat and sorghum ground where margins are repeatedly weak or negative, soils are degraded, and continued cultivation accelerates erosion without generating reliable income. Class A acres receive the full stewardship payment (\$225/acre) as permanent or long-term retirement from row-crop production. Estimated scope: 15–20 million acres.

3.2 Class B: Working Grassland Conversion

These acres transition to managed grazing operations using bison, cattle, or mixed herds under rotational protocols that mimic historical grazing patterns. Compatible with Grassland CRP-like models, Class B enrollment maintains agricultural identity and cultural continuity while restoring grass cover. Payment levels are lower (\$100–150/acre base) but supplemented by livestock revenue, seed harvest, and hunting leases. Estimated scope: 20–25 million acres.

3.3 Class C: Hydrologic and Biodiversity Priority Land

These are the acres that matter most for water security and ecological connectivity: playas, riparian corridors, wetland margins, erosion-prone slopes, and habitat connectors between fragmented prairie remnants. Class C enrollment receives targeted payments reflecting the high ecosystem-service value of these specific landscapes. Playa protection is especially critical: USGS research demonstrates that recharge beneath playas can be one to two orders of magnitude higher than in surrounding interplaya settings. Class C acres are the highest priority for permanent conservation easement. Estimated scope: 5–10 million acres.

This classification serves multiple strategic purposes. It lowers the aggregate fiscal burden compared to a uniform premium across all 50 million acres. It aligns with USDA working-lands program philosophy for Class B. It focuses hydrological claims on the specific landscapes where science supports them most strongly. And it avoids the perception of a culturally insensitive program that seeks to empty the Plains—Class B enrollment keeps ranchers ranching, on better grass, with lower risk.

4. Hydrological Benefits and Water Security

4.1 The Primary Benefit: Reduced Withdrawals

The most impactful hydrological benefit of large-scale prairie restoration is reduction in irrigation demand. Irrigated agriculture accounts for approximately 70% of Ogallala Aquifer withdrawals. Converting marginal irrigated acres to prairie eliminates their water demand entirely. Converting dryland acres reduces the political and economic pressure to expand irrigation into new areas. At a landscape level, the transition creates a net positive water budget in many counties for the first time in decades.

This is the strongest and most defensible hydrological claim: prairie restoration helps the aquifer primarily by stopping the withdrawals that are depleting it, rather than by directly recharging it.

4.2 Soil and Surface Water Improvements

Prairie restoration creates measurable improvements to soil structure, water infiltration, and surface hydrology. Deep prairie root systems (reaching 15 feet or more) create macropores that increase water infiltration rates relative to compacted cropland. Restored prairies reduce flashy runoff patterns that cause flooding and erosion. Tributaries feeding into major river systems experience more consistent baseflows. Soil erosion into waterways decreases by an estimated 80–95%. Agricultural chemical inputs to surface water decline dramatically.

4.3 Aquifer Recharge: A Nuanced Picture

The relationship between land cover and aquifer recharge is more complex than earlier formulations suggested. Recharge rates in the High Plains are spatially heterogeneous, driven by geology, soil type, topography, and the presence or absence of playas—shallow, seasonal wetlands that serve as critical recharge windows.

USGS research on the High Plains aquifer emphasizes that recharge beneath playas can be one to two orders of magnitude higher than recharge in surrounding interplaya settings. This finding has important implications for restoration design: protecting and restoring playas (Class C priority acres) may deliver more recharge benefit than converting vast acreage of upland cropland. In some upland settings, research has found that land-cover changes produce little detectable difference in recharge rates, or that the direction of effect depends on crop type, soil, and climate.

The scientifically defensible position is this: prairie restoration improves hydrologic resilience primarily by reducing pumping, restoring soil structure, reducing runoff and erosion, protecting playas and riparian buffers, and improving watershed function. Direct enhancement of deep aquifer recharge is an additional potential benefit that varies by landscape position and is strongest in playa-associated settings.

On recovery timelines: the Ogallala Aquifer would require approximately 6,000 years to fully replenish naturally. The realistic goal is not recovery but stabilization—halting the crisis of depletion and establishing a sustainable water balance. Even modest stabilization within 10–15

years of large-scale restoration would represent a transformative reversal of current trends and buy time for longer-term solutions.

4.4 Climate Feedback Effects

Increased soil moisture from restored prairies creates measurable localized cooling effects of 2–4°F. Enhanced evapotranspiration moderates temperature extremes and increases regional humidity. Research indicates potential increases in localized precipitation of 5–10% through improved moisture recycling. These combined microclimate improvements benefit adjacent agricultural lands that remain in production.

5. National Security and Strategic Autonomy

This is the section that did not exist in the original proposal. Events since May 2025 have made it the strongest argument in the paper.

The United States' current agricultural model on the Great Plains is built on a supply chain that runs through some of the most geopolitically unstable corridors on Earth. Nitrogen fertilizer depends on natural gas, much of which is priced against global markets disrupted by the wars in Ukraine and the Middle East. Potash flows through supply chains linked to Russia, Belarus, and Canada. Phosphate involves Morocco, China, and conflict-adjacent logistics. Diesel fuel—the lifeblood of mechanized agriculture—is subject to Hormuz chokepoint risk.

Prairie stewardship acreage is insulated from all of these chokepoints. It requires no fertilizer, no pesticides, no diesel for field operations, and no seed purchases. It produces economic value (stewardship payments, grazing revenue, ecosystem services) from inputs that are entirely domestic: soil, rain, sunlight, and native plant genetics. In an era of supply-chain fragility, a strategic reserve of low-input landscape is not a luxury—it is a national security asset.

The program can therefore be framed as:

- Lowering American agriculture's exposure to foreign fertilizer chokepoints.
- Reducing diesel dependence on marginal acres that don't generate reliable returns even with cheap fuel.
- Stabilizing rural household income under wartime price shocks.
- Treating restored prairie not just as habitat but as strategic reserve landscape—acreage that can be returned to production in a true food-security emergency but that generates more value as prairie in normal conditions.

This framing has bipartisan appeal that transcends the climate-versus-economy divide. It speaks to national defense, supply-chain resilience, rural self-sufficiency, and American strategic autonomy—values that resonate across the political spectrum.

6. Implementation Strategy and Timeline

6.1 Regional Phasing

The initiative follows a three-phase regional implementation plan designed to maximize visibility, ecological connectivity, and stakeholder engagement.

Phase 1: First Islands (Years 1–3)

Central Kansas Smoky Hills: 2–3 million acres. Contains fragmented tallgrass remnants, severe Ogallala depletion, moderate agricultural profitability. Creates connectivity between Tallgrass Prairie National Preserve and Konza Prairie.

Nebraska Sand Hills Expansion: 3–4 million acres. Adjacent to the largest remaining intact prairie system. Compatible with existing ranching operations (primarily Class B enrollment).

Oklahoma-Texas Panhandle Junction: 2–3 million acres. Severe groundwater depletion, high climate vulnerability. Demonstrates cross-state collaboration and the Kansas July 2026 water-management mandate.

Phase 2: Major Expansion (Years 3–7)

Dakota Mixed-Grass Corridor: 8–10 million acres across eastern North and South Dakota. Creates critical habitat corridor and establishes a major continuous carbon sink.

Western Great Plains Shortgrass Mosaic: 10–12 million acres across eastern Colorado, eastern Wyoming, and western Nebraska. Drought-prone regions with marginal agriculture and high wind erosion potential.

Southern Plains Expansion: 8–10 million acres in western Oklahoma and the Texas Panhandle. Showcases transition to sustainable ranching practices.

Phase 3: Connectivity and Completion (Years 7–10)

The final phase fills gaps between established restoration zones (6–8 million acres), develops the eastern prairie-forest transition zone (5–7 million acres), and completes the western range boundary (3–5 million acres). When complete, the initiative will have created the world's largest restored ecosystem and a continuous north-south wildlife corridor spanning from the Dakotas to the Texas Panhandle.

6.2 Strategic Seedstock Production

A critical innovation of this proposal is using Phase 1 restoration islands as seedstock generators for subsequent expansion. Ten to fifteen percent of each Phase 1 island is designated specifically for native seed production, with cooperative equipment pools and processing facilities. This creates a self-reinforcing feedback loop: early success produces the biological inputs needed for later phases, addressing the native seed supply constraint that has limited previous large-scale restoration efforts.

6.3 Monitoring and Verification

The initiative employs satellite-based remote sensing (Landsat, Sentinel-2) as its primary monitoring platform, with AI algorithms trained to detect prairie biosignatures that distinguish restored prairie from cropland. Strategic ground-truthing validates remote sensing data. Key performance indicators include: acres converted, plant community composition and diversity, soil carbon accumulation, water infiltration improvements, wildlife population response, and economic indicators including job creation and tourism revenue. Independent third-party verification of carbon sequestration and transparent public reporting ensure accountability.

7. Ecological Restoration and Wildlife Recovery

Large-scale prairie restoration will catalyze a wildlife renaissance across the Great Plains. The creation of continuous habitat corridors at landscape scale—rather than isolated conservation fragments—enables ecological processes that small preserves cannot support.

7.1 Habitat Quality and Corridor Creation

The primary wildlife benefit is habitat quality and connectivity at a scale sufficient to support viable populations of area-sensitive species. Grassland birds—among the most rapidly declining bird guilds in North America, with over 40 species affected—will gain the large, contiguous habitat blocks they require for breeding success. Pollinator communities will benefit from the diverse native flora of restored prairie, with monarch butterflies and native bees among the most prominent beneficiaries. Pronghorn may expand their range eastward through newly created corridors. Stream and riparian restoration embedded within the prairie matrix will improve habitat for native fish species and moderate stream temperatures.

7.2 Keystone Species and Recreation

Strategic reintroduction of bison to appropriate Class B and Class C areas will restore the keystone grazing influence that historically shaped prairie ecology, creating habitat heterogeneity that benefits numerous dependent species while providing economic opportunities through meat production and tourism. Managed hunting programs for upland game birds—prairie chickens, sharp-tailed grouse, and pheasants—will create world-class hunting opportunities as populations recover in response to restored habitat. A managed-access program through the Cooperative will provide high-quality, low-density hunting experiences while generating revenue for participating landowners.

Wildlife viewing infrastructure, including the proposed “Prairie Passage” long-distance trail system connecting major restoration areas, will support a growing nature-based tourism sector.

8. Governance: The Prairie Restoration Cooperative

8.1 Why a Cooperative, Not a Federal Program

The governance design has been substantially revised in this edition to reflect a hard lesson from 2025–2026: federal conservation infrastructure cannot be assumed stable across administrations. USDA lost 27% of its workforce in a single year. NRCS—the agency most relevant to this proposal—lost 22% of staff, including experienced soil scientists, hydrologists, and conservation planners whose expertise takes years to develop. A 50-million-acre, multi-decade initiative cannot be built on institutional foundations that lose a quarter of their capacity when political winds shift.

The Prairie Restoration Cooperative is therefore designed as a state-chartered, farmer-owned entity with independent technical staff, modeled on the Farm Credit System and rural electric cooperatives—institutions that have demonstrated remarkable durability across political cycles precisely because they are member-owned and operationally autonomous.

8.2 Cooperative Structure

- Chartered under state cooperative law in each participating state, with a federal coordinating overlay for interstate programs.
- Automatic membership for participating landowners with voting rights proportional to enrolled acreage.
- Board composition: landowner representatives (60%), scientific and technical advisors (20%), public interest representatives (20%).
- Regional chapters aligned with implementation phases.
- Independent technical staff: the Cooperative hires its own soil scientists, hydrologists, restoration ecologists, and fire management specialists—building the technical capacity that can no longer be assumed from NRCS.
- Authority to issue bonds backed by payment streams, enter long-term contracts (15+ years), aggregate and market ecosystem services, and coordinate with federal programs when staff and funding are available.

8.3 Political Durability

The Cooperative creates its own constituency of member-owners with direct economic interest in program continuation. It is less vulnerable to administration changes than a pure government program. It maintains operational continuity independent of federal staffing levels. And it can receive funding from multiple sources—federal, state, private, and market-based—without being dependent on any single appropriation.

9. Funding: A Layered Finance Model

9.1 Beyond CRP

This initiative is not a scaled-up Conservation Reserve Program. CRP is capped at 27 million acres by statute and is nearly full, with only 1.9 million acres available for new enrollment in fiscal year 2026. CRP's average national rental payment is approximately \$72/acre—well below the

stewardship payments proposed here. And CRP was not designed for the strategic, multi-class landscape transformation this proposal envisions.

Instead, the Cooperative functions as a strategic overlay that aggregates multiple funding streams into a unified producer contract:

Federal conservation programs: CRP, Grassland CRP, EQIP, CSP, ACEP, and the new Agriculture Management Practices (AMP) initiative all provide relevant funding that can be stacked and coordinated. The key principle from AMP—that at least 65% of federal funds go directly to producers—aligns with the Cooperative’s payment-first design.

State water-security appropriations: Kansas’s new groundwater management mandate creates a direct state interest in funding reduced withdrawals. Similar state-level water programs exist or are emerging in Texas, Oklahoma, Colorado, and Nebraska.

Bond finance: The Cooperative’s authority to issue bonds backed by aggregated payment streams provides capital for upfront restoration costs and infrastructure development.

Ecosystem-service markets: Carbon sequestration credits, water-quality improvement credits, and biodiversity credits provide long-term revenue streams that grow as restored prairies mature and markets develop.

Philanthropic and impact investment: Conservation-oriented foundations, impact investors, and private-sector partners in tourism infrastructure provide supplementary funding.

9.2 Budget Estimate

At the proposed payment structure across three acreage classes, the aggregate annual cost at full implementation (50 million acres) is approximately \$5.5–7.5 billion, depending on the Class A/B/C distribution. This figure is substantial but contextualizable: USDA currently spends \$8–10 billion annually on crop insurance subsidies, much of it supporting the same marginal acres that this program would transition to more productive use. Redirecting a fraction of existing agricultural support—combined with state contributions, bond proceeds, and ecosystem-service revenue—can capitalize the program without requiring major new federal appropriations.

10. Legal Considerations

10.1 Water Rights

Water law in the Great Plains varies significantly by state and directly affects the feasibility and design of the initiative. Texas follows a “right of capture” doctrine, meaning groundwater belongs to the landowner and can be pumped without regard to impact on neighbors—which paradoxically makes voluntary retirement of water rights more difficult to enforce, since a landowner enrolled in prairie stewardship cannot prevent a neighbor from pumping the same aquifer more aggressively. Kansas follows correlative rights principles and has established groundwater management districts with statutory authority to restrict pumping—a more favorable legal framework for coordinated

withdrawal reduction. Nebraska’s system of natural resources districts provides another model. The Cooperative’s contract design must be tailored to each state’s water-law framework, with particular attention to whether enrolled acres can secure legally enforceable water-right retirements.

10.2 Conservation Easements and Land-Use Restrictions

Permanent conservation easements on Class C priority acres require compliance with state easement law, including requirements for qualified holders, baseline documentation, and monitoring obligations. The federal tax deduction for conservation easement donations (under IRC §170(h)) provides an additional financial incentive for landowners who choose permanent protection. However, the Cooperative must ensure that easement terms are consistent across state lines and that monitoring obligations can be sustained over multi-decade timeframes.

10.3 CRP Acreage Cap as Legislative Barrier

The 27-million-acre statutory cap on CRP enrollment is a binding constraint. Legislation to raise the cap—or, preferably, to create a new strategic landscape contract authority distinct from CRP—will be necessary for the initiative to reach its full 50-million-acre scope. The pending Farm, Food, and National Security Act of 2026 provides a potential legislative vehicle, and the Cooperative should engage in the Farm Bill negotiation process to advocate for expanded conservation acreage authority.

10.4 Prescribed Fire Liability

Prescribed fire is essential for long-term prairie health but carries legal liability risk. States vary in their treatment of prescribed fire liability, from strict liability (landowner bears all risk) to certified-burner protections that limit liability for trained practitioners following approved burn plans. The Cooperative will need to develop cooperative-wide liability coverage, invest in certified-burner training programs, and advocate for state-level prescribed-fire liability reform where existing law creates undue barriers.

10.5 Tribal Sovereignty and Consultation

Several proposed restoration areas overlap with or adjoin tribal lands. The Cooperative must engage in government-to-government consultation with affected tribal nations, respect treaty rights and sovereignty, and explore partnership models that support tribal land management objectives. Tribal nations have deep ecological knowledge of prairie systems and should be partners, not afterthoughts, in the restoration design process.

11. Acknowledging Transition Impacts

A credible policy proposal names its costs plainly. The transition from input-intensive cropping to prairie stewardship will create losers as well as winners, and addressing these impacts honestly is essential for political viability.

Fertilizer and chemical dealers: Reduced demand for synthetic fertilizers, herbicides, and pesticides will contract revenue for agricultural input suppliers. The Cooperative should support transition assistance for these businesses, including retraining programs and diversification support.

Farm equipment dealers: Reduced equipment needs on enrolled acres will lower demand for new tractors, planters, and combine harvesters. However, new equipment needs will emerge for restoration-specific machinery (seed drills, prescribed fire equipment, grazing infrastructure).

Custom applicators and crop consultants: Businesses that depend on input-intensive agriculture for their client base will see reduced demand. The Cooperative's technical staffing needs create partial offset.

County tax bases: Changes in land use may affect property tax assessments. State legislation may be needed to ensure that enrolled prairie acres receive appropriate tax treatment that neither penalizes participating landowners nor destabilizes county revenue.

Grain elevator and processing infrastructure: Reduced grain volume from enrolled acres affects downstream infrastructure. However, Phase 1 islands are specifically located in regions with the weakest agricultural margins, where elevator utilization is already declining.

The proposal addresses these transition impacts through phased implementation (allowing gradual adjustment), development of new prairie-based enterprises (seed production, ecotourism, managed grazing), and explicit inclusion of community-development payments (10% of the payment structure) for affected rural businesses and infrastructure.

12. Risk Management Framework

12.1 Ecological Risks

Invasive species: Early detection through satellite monitoring combined with ground surveys; rapid response protocols; integrated management using strategic mowing, targeted grazing, selective herbicide, and biological controls; rigorous seed-source testing.

Fire management: Scientifically-based prescribed fire program with certified practitioner training; regional fire councils; cooperative-wide liability coverage; public education on ecological necessity of fire.

Drought and climate resilience: Diverse seed mixes including drought-adapted ecotypes; adaptive planting timing; contingency funding for extreme events; climate-specific strategies by region.

12.2 Political and Economic Risks

Federal funding volatility: The Cooperative's multi-source funding model and bond-issuance authority reduce dependence on any single appropriation. The 2025–2026 USDA staffing crisis validates this design choice.

Administration changes: State charter, member ownership, and operational autonomy provide institutional continuity independent of federal political cycles.

Payment reserve: A financial reserve fund ensures payment continuity regardless of federal budget fluctuations.

Market development: Active creation of markets for prairie products (native seed, grazing, carbon credits, tourism) ensures long-term economic sustainability beyond federal payments.

13. Expected Outcomes

13.1 Environmental

- Restoration of native tallgrass and shortgrass prairie across 50 million acres—the world’s largest restored ecosystem.
- Creation of continuous north-south wildlife corridor from the Dakotas to Texas.
- Recovery of declining grassland bird populations and pollinator communities.
- Conservative annual carbon sequestration of 37.5 million metric tons CO₂ (0.75 t/acre × 50M acres).
- Significant improvement in regional water quality, watershed function, and playa protection.
- Aquifer stabilization in critical depletion zones through reduced withdrawals and enhanced infiltration.

13.2 Economic

- Stable, predictable income for tens of thousands of agricultural landowners.
- Creation of 25,000–35,000 jobs in restoration, seed production, grazing management, and tourism.
- Development of a \$1.5–2 billion annual native seed industry.
- Growth of a \$3–5 billion annual nature-based tourism sector.
- Reduced federal expenditures on disaster assistance and crop insurance for marginal acres.

13.3 Strategic

- Reduced national dependence on foreign-sourced fertilizer and fuel for food production.
- Strategic landscape reserve available for emergency production if needed.
- Model for large-scale public-private partnership in landscape restoration.
- Bipartisan demonstration that environmental restoration and rural economic development are the same project.

14. Conclusion

The Re-Greening the Heartland Initiative represents a once-in-a-generation opportunity to transform the American landscape while revitalizing rural communities, securing water resources, and strengthening national strategic autonomy. The convergence of fertilizer chokepoints, fuel price volatility, commodity oversupply, aquifer depletion, and federal conservation infrastructure collapse has made the case for this initiative substantially stronger than when it was first proposed in 2025.

The American heartland once supported the world's greatest grassland ecosystem. With vision, determination, and a governance structure designed for durability rather than political convenience, we can restore a significant portion of this natural heritage while creating a more prosperous, resilient, and strategically autonomous future for the communities that steward these lands.

The time to act is now. The economic, ecological, strategic, and water-security imperatives have aligned to create this moment of opportunity. Every year of delay is another foot of aquifer lost, another generation of farm families driven from the land, and another billion dollars spent subsidizing an agricultural model that enriches input suppliers while impoverishing the people who work the soil.

This proposal offers a different path. It is ambitious but practical, visionary but grounded in scenario economics and legal realism, and designed to endure across the political cycles that have undermined every previous attempt at landscape-scale conservation in American history.

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